



B.3 Gas laws

The particulate nature of matter · IB Diploma · Concept guide and original exam-style practice

Understand the concept

An ideal gas obeys $pV = nRT$ (or $pV = NkT$ using number of particles N and Boltzmann constant k), linking pressure p , volume V , amount of substance n (or particle number N), and absolute temperature T , with R the molar gas constant. The kinetic model of an ideal gas assumes point-like particles in constant random motion undergoing perfectly elastic collisions with no intermolecular forces except during collisions; this model predicts $pV = \frac{1}{3}Nm\langle c^2 \rangle$, where m is particle mass and $\langle c^2 \rangle$ the mean square speed, and shows that average translational kinetic energy per particle, $\langle E_k \rangle = \frac{3}{2} kT$, is directly proportional to absolute temperature. For a fixed amount of gas undergoing a change of state, $p_1V_1/T_1 = p_2V_2/T_2$.

Worked example: follow the reasoning

A gas occupies $2.0 \times 10^{-3} \text{ m}^3$ at 300 K and $1.0 \times 10^5 \text{ Pa}$. It is compressed to $1.2 \times 10^{-3} \text{ m}^3$ and heated to 400 K. Find the new pressure.

$$p_1V_1/T_1 = p_2V_2/T_2$$

$$(1.0 \times 10^5)(2.0 \times 10^{-3})/300 = p_2(1.2 \times 10^{-3})/400$$

$$666.7/300 \dots \text{compute directly: } p_2 = p_1V_1T_2/(T_1V_2) = (1.0 \times 10^5)(2.0 \times 10^{-3})(400)/(300 \times 1.2 \times 10^{-3})$$

$$= (80)/(3.6 \times 10^{-1}) = 80/0.36 = 222.2 \times 10^3 \dots \text{let's redo carefully:}$$

$$\text{Numerator} = 1.0 \times 10^5 \times 2.0 \times 10^{-3} \times 400 = 8.0 \times 10^4.$$

$$\text{Denominator} = 300 \times 1.2 \times 10^{-3} = 0.36.$$

$$p_2 = 8.0 \times 10^4 / 0.36 = 2.22 \times 10^5 \text{ Pa.}$$

$$\text{Answer: } p_2 \approx 2.2 \times 10^5 \text{ Pa.}$$

Quick check · multiple choice

According to the kinetic model of an ideal gas, what happens to the average kinetic energy of the gas particles if the absolute temperature is doubled?

- A Stays the same
- B Doubles
- C Quadruples
- D Halves

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 An ideal gas sample of 0.040 mol is held in a sealed container at a pressure of $1.5 \times 10^5 \text{ Pa}$ and a temperature of 290 K. [7]
 - (a) Calculate the volume of the gas using $pV = nRT$ ($R = 8.31 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$).
 - (b) The gas is heated at constant volume until its pressure rises to $2.0 \times 10^5 \text{ Pa}$. Calculate the new temperature.
 - (c) Explain, in terms of the kinetic model, why raising the temperature at constant volume increases the pressure.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $V = nRT/p = (0.040)(8.31)(290)/(1.5 \times 10^5) = 96.4/1.5 \times 10^5 = 6.43 \times 10^{-4} \text{ m}^3$.

(b) Constant volume: $p_1/T_1 = p_2/T_2 \rightarrow T_2 = T_1(p_2/p_1) = 290 \times (2.0 \times 10^5/1.5 \times 10^5) = 290 \times 1.333 = 386.7 \text{ K} \approx 387 \text{ K}$.

(c) Higher temperature increases the average kinetic energy and hence the speed of gas particles; they collide with the container walls more frequently and with greater momentum change per collision, increasing the average force per unit area, i.e. the pressure.

Quick check answer

B. Average translational kinetic energy $\langle E_k \rangle = 3/2 kT$ is directly proportional to absolute temperature T , so doubling T doubles $\langle E_k \rangle$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.